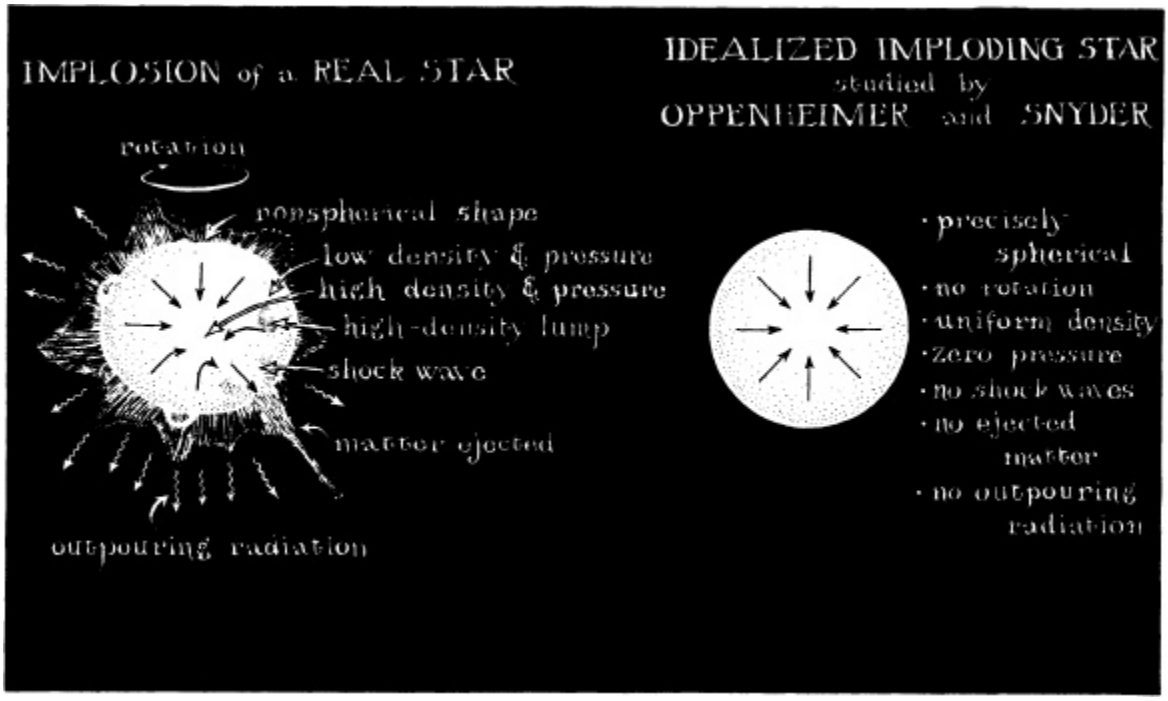
claimed, *these deformations will grow large and halt the implosion before a singularity can form*. Similarly, Khalatnikov and Lifshitz asserted, random deformations will halt the big crunch implosion of our entire Universe eons hence, and thereby save the Universe from destruction in a singularity.

Khalatnikov and Lifshitz came to these views in 1961 by asking themselves whether, according to Einstein's general relativistic laws, singularities are *stable against small perturbations*. In other words, they posed the same question for singularities as we met in Chapter 7 for black holes: If, in solving Einstein's field equation, we alter, in small but random ways, the shape of the imploding star or Universe and the velocity and density and pressure of its material, and if we insert into the material tiny but random amounts of gravitational radiation, how will these changes (these *perturbations)* affect the implosion's predicted endpoint?

For the black hole's horizon, as we saw in Chapter 7, the perturbations make no difference. The perturbed, imploding star still forms a horizon, and although the horizon is deformed at first, all its deformations quickly get radiated away, leaving behind a completely "hairless" black hole. In other words, the horizon is *stable* against small perturbations.



13.2 (Same as Figure 6.3.) ***Left:*** **Physical phenomena in a realistic, imploding star.** ***Right:*** **The idealizations which Oppenheimer and Snyder made in order to compute stellar implosion. For a detailed discussion see Chapter 6.**